

Designing for Staying with Emotions: An AR System Driven by Generative AI to Support Emotional Participation in Children with Autism

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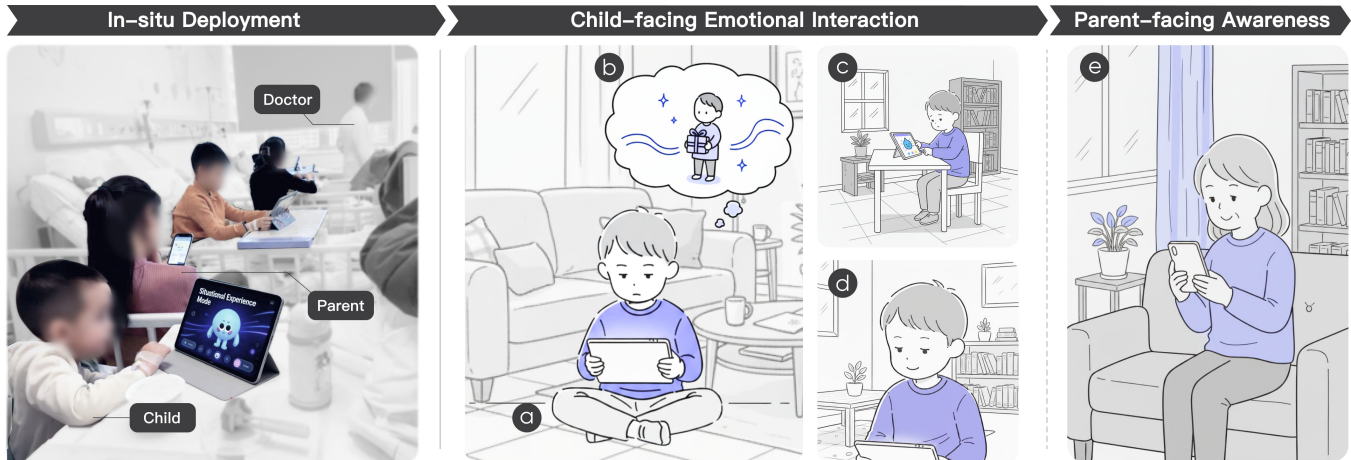


Figure 1: MOODMATE system overview and user scenarios: (a) in-situ deployment with child, parent, and clinician; (b) child in daily context; (c) system-mediated emotional interaction; (d) gentle emotional guidance; (e) parent-facing awareness interface.

Abstract

Children with autism spectrum disorder (ASD) often face challenges in emotion recognition and expression. Existing emotional intervention approaches remain limited in their ability to provide personalized support, facilitate contextual transfer, and offer adaptive interaction feedback. Recent advances in generative artificial intelligence and augmented reality (AR) have opened new possibilities for designing situated and immersive emotional support systems; however, systematic explorations of their interaction design for neurodiverse children remain scarce. In this paper, we present MOODMATE, a multimodal emotional cognition support system that integrates generative AI and AR. The system supports children with ASD in engaging with emotional understanding and expression within real-life contexts through character-driven situated narratives, parent-involved personalization mechanisms, and multimodal emotional feedback. We report findings from a four-week exploratory deployment study in which ten children with ASD aged 5–10 used the system in their home environments. Drawing on scale-based measures, system usage logs, and parent interviews,

we identify patterns of multimodal emotional interaction, experiential characteristics, and emerging design challenges. Our findings highlight design opportunities related to safety, interpretability, and parental collaboration in generative AI-driven emotional support systems, and offer preliminary insights for the interaction design of emotional interventions for neurodiverse children.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI); Interactive systems and tools; User interface toolkits;**

Keywords

Autism Spectrum Disorder, Generative Artificial Intelligence, Multimodal Interaction, Augmented Reality, Emotional Cognition Support



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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798902>

ACM Reference Format:

Yue Yu and Zhitao Yu. 2026. Designing for Staying with Emotions: An AR System Driven by Generative AI to Support Emotional Participation in Children with Autism. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798902>

1 Introduction

Children with autism spectrum disorder (ASD) often experience challenges in emotion recognition and expression [15], making emotional support a longstanding concern across education, psychology, and human–computer interaction research [1, 4, 5]. Emotional understanding is widely regarded as a foundation for social interaction and emotional regulation [7, 9, 22], directly influencing children’s ability to participate in everyday social situations. As a result, supporting emotional development in children with ASD has remained an important focus of both research and practice.

In existing practice, approaches such as Applied Behavior Analysis (ABA) and Social Skills Training (SST) have been widely adopted to help children with ASD learn emotional cues and social rules [13, 26, 27]. These methods typically rely on highly structured procedures and adult-led guidance, with interventions conducted in controlled or specialized settings. While such approaches can be effective in constrained contexts [11, 23, 24], they are often difficult to sustain in children’s everyday lives and may not adequately accommodate the dynamic nature of emotional experiences or the non-verbal and individualized ways in which children with ASD express emotions [8]. This limitation highlights an ongoing gap in emotional interventions with respect to contextual continuity and individual adaptability.

With the development of digital technologies, researchers have begun to explore multimodal interactive systems and augmented reality (AR) as means to provide more engaging and contextually grounded emotional support. Multimodal systems combine speech, behavior, and interaction cues to support richer forms of emotional expression [6, 21, 28], while AR embeds virtual elements into physical environments [12, 19, 20, 30], offering interaction experiences that are more closely situated in everyday life [3, 17, 18, 29]. In parallel, recent advances in generative artificial intelligence have enabled more flexible forms of emotional feedback beyond pre-defined scripts [12, 19, 20]. However, when applied to children with ASD, these technologies also introduce new design challenges. These include how to provide interpretable and emotionally safe feedback given fragmented and non-verbal emotional expressions, how to prevent emotional misguidance while maintaining generative flexibility, and how to incorporate parental involvement over long-term use to support reliable emotional understanding and regulation. Systematic design explorations addressing these challenges remain limited.

To address this gap, we present MOODMATE, a multimodal emotional cognition support system that integrates generative artificial intelligence and augmented reality to explore new approaches to emotional interaction design for children with ASD. MOODMATE embeds emotional support into children’s everyday contexts through character-driven situated narratives, multimodal emotional feedback, and parent-involved regulation mechanisms, allowing interactions to evolve in response to both contextual conditions and children’s ongoing engagement. Building on this system, we conducted a four-week exploratory deployment study in home environments to examine how MOODMATE was used in practice. Our findings characterize patterns of participation, experiential features, and interactional changes in multimodal emotional engagement, and highlight key design considerations for emotional interaction

systems for neurodiverse children, including interpretability of emotional feedback, adaptability of interaction rhythm, and the role of parental collaboration in family settings.

2 Moodmate: An Exploratory System

2.1 Design Motivation and Strategy

MOODMATE was designed as an exploratory system to investigate how generative artificial intelligence and augmented reality (AR) can support emotional interaction for children with autism spectrum disorder (ASD) in everyday life contexts. Rather than functioning as an intervention tool oriented toward training effectiveness, the system focuses on how emotional support is constructed, experienced, and gradually adjusted through real-world use, particularly with respect to sustainability and adaptability in long-term home environments.

The system design was guided by three core strategies. First, emotional interaction was conceptualized as a situated and gradual experience rather than a one-time task or assessment. Second, the system was designed to support diverse and non-verbal forms of emotional expression, reducing reliance on children’s verbal abilities. Third, emotional feedback needed to remain interpretable and allow parental involvement in the regulation process. Based on these strategies, MOODMATE was developed as an evolving interactive system intended to explore the design possibilities and tensions associated with generative emotional feedback, multimodal input, and family collaboration in real-world use.

2.2 System Architecture Overview

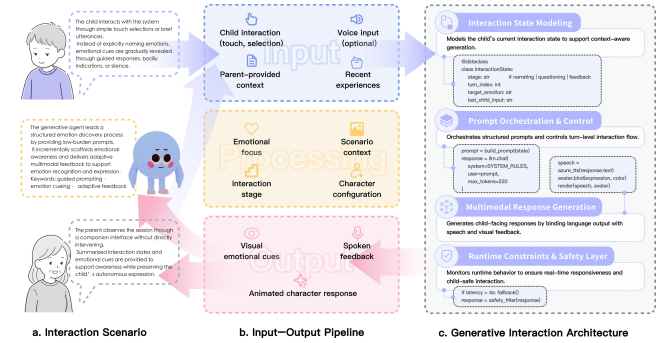


Figure 2: MOODMATE system overview: (a) interaction scenario in home deployment; (b) input-output pipeline; (c) generative interaction architecture.

The MOODMATE system consists of three interrelated components: a child-facing AR interaction interface, a generative AI-based emotional feedback module, and a parent-facing collaborative regulation interface. Together, these components support sustained emotional interaction within home environments.

On the child side, mobile AR technology overlays virtual characters onto the physical environment, guiding children into emotion-related situations and supporting interaction. The generative AI module dynamically produces feedback based on children’s multimodal inputs during interaction, allowing emotional support to

adapt to changing contexts. The parent-facing interface enables caregivers to review emotional records, supplement contextual observations, and adjust interaction pacing, thereby forming a mechanism for family collaboration during system use.

This architecture emphasizes coordination among components rather than the performance of individual modules, allowing the system to function as an integrated whole that supports emotional interaction over time.

2.3 Situated Interaction and Character Design

MOODMATE’s interaction design is grounded in everyday life contexts. Instead of static tasks or standardized training flows, the system guides children through a series of situated “micro-narratives” that encourage emotional engagement [14, 25, 31]. Each scenario typically includes three stages: contextual introduction, emotional triggering, and guided response, allowing children to gradually attend to emotional cues within familiar settings.

To accommodate the perceptual and cognitive characteristics of children with ASD [2, 12], the system incorporates three geometric emotional interaction characters—Bubble, Corner, and Supple. The characters follow low-complexity and high-symmetry design principles and employ cool color palettes with high brightness to reduce perceptual load while supporting sustained attention [2, 10, 16]. Emotional information is conveyed through color changes, simple facial expressions, voice prompts, and lightweight haptic feedback, ensuring that emotional feedback remains intuitive and accessible.

Throughout interaction, the system supports multiple non-verbal participation modes. Children may express emotions through pointing, selection, or simple actions, thereby reducing the dependence of emotional interaction on verbal expression.

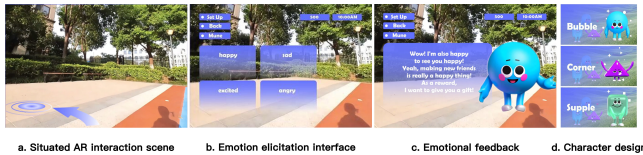


Figure 3: In-situ AR emotional interaction with MOODMATE: (a) situated AR interaction scene; (b) emotion elicitation interface; (c) emotional feedback; (d) character design.

2.4 Multimodal Input and Feedback Loop

MOODMATE incorporates multimodal inputs to support the interpretation of children’s emotional states, including vocal features, interaction behaviors, and task response patterns. During interaction, these inputs are integrated to form an initial, situated understanding of the child’s current emotional state, which then informs the generation of corresponding feedback. To ensure interpretability and consistency in emotional interaction, the system does not rely on fully open-ended dialogue generation. Instead, it adopts a structured emotional dialogue framework that differentiates the functional roles of various prompts within each interaction cycle.

Emotional feedback follows a gradual guidance strategy. Rather than providing immediate evaluation or correction, the system encourages continued engagement through prompts, encouragement,

Table 1: Structure of the MOODMATE emotional dialog

Sentence Types	Functional Definition	Example (generated by GPT-4o)
descriptive sentence	Provide context for emotional events	“Today I was ostracized on the playground and I felt a little sad.”
transport and sentences	Eliciting empathy or emotional associations	“Have you ever experienced this feeling?”
command sentence	Encouragement of target behaviors	“If I were you, I’d try to tell the teacher how I feel.”
question period	Detecting Emotional Understanding	“What mood do you think I’m in?”
affirmative sentence	Provide encouraging feedback	“That’s a good idea! I really like your answer.”

and illustrative examples. Parents are able to review emotional records during system use and supplement or adjust emotional interpretations in specific contexts. When parental input is provided, the system adapts the presentation of subsequent interaction content, forming a closed loop of interaction, feedback, and adjustment.

This feedback loop is not designed to optimize a predefined performance metric. Instead, it serves to surface variations in use and emerging design tensions across different family contexts, providing a basis for the system’s ongoing evolution.

3 User Study

To explore how MOODMATE was used and experienced in real-world home settings, we conducted an exploratory user study. Ten children with ASD aged 5–10 and their families were recruited, and the system was deployed in participants’ homes for four weeks. During the study, parents guided children’s interaction with the system without imposed usage frequency or fixed task requirements, allowing engagement to align with children’s daily states and routines.

The system automatically logged interaction data and emotional feedback throughout the deployment. Following the study, semi-structured interviews were conducted with parents, three special education teachers, two psychiatrists, and one child psychological counselor. Data analysis primarily employed qualitative methods, focusing on how children participated in situated emotional interaction, how system design shaped emotional expression, and how parental collaboration influenced emotional support processes, rather than evaluating quantitative changes in emotional intervention outcomes.

4 Results

Based on system logs and semi-structured interviews collected during the four-week home deployment, we identified three core findings. These findings reveal how children with ASD engaged in

emotional interaction in AR-mediated home contexts, as well as key interactional characteristics that emerged during everyday use.

Diverse Patterns of Emotional Participation. Children demonstrated noticeably different patterns of emotional participation when interacting with MOODMATE. Some children showed sustained visual attention to the AR character as it appeared in their physical environment, particularly responding to changes in color, form, and presence, and engaged through repeated interactions. Others participated primarily through brief actions, such as selecting options or pointing, with minimal reliance on verbal expression. By embedding emotional interaction into the children's lived environment, AR enabled multiple modes of participation without requiring a uniform expressive pathway. These observations suggest that emotional participation was closely aligned with individual perceptual and expressive preferences, rather than being determined by a predefined interaction flow.

Structured Dialogue Stabilized Interaction Rhythm. The structured dialogue design provided children with a predictable interaction rhythm, helping to reduce uncertainty during emotional engagement. In the AR setting, this structure was not only conveyed through language but also made perceptible through the character's appearance, persistence, and timing of feedback. Parents reported that such rhythmic consistency supported children's continued participation, even when verbal expression was limited. Importantly, the dialogue structure did not guide children toward correct or expected responses; instead, it functioned as an interactional scaffold that stabilized emotional engagement.

Parental Involvement as Ongoing Contextual Calibration. Parents did not act merely as supervisors during system use, but participated as contextual calibrators by supplementing background information and adjusting interaction pacing. The AR-based presentation helped parents relate system feedback to children's immediate life situations, allowing emotional interaction to remain grounded in everyday contexts. At the same time, differences in parental involvement across households highlighted variations in how emotional support was enacted, pointing to underlying design tensions in family-mediated emotional systems.

5 Discussion and Conclusion

This study explored how generative AI combined with augmented reality can support emotional interaction for children with ASD in home settings. Rather than framing emotional interaction as a training task, this work focused on how emotions are experienced, entered, and stayed with through embodied and situated interactions.

From Emotional Training to Staying with Emotions. Our findings indicate that children's engagement with MOODMATE was not primarily reflected in immediate improvements in emotion recognition or labeling. Instead, participation often manifested as remaining with emotional situations through observation, action, or partial responses. By situating emotions within the physical environment through AR, emotions became experiential contexts rather than abstract conversational objects. This shift suggests an alternative approach to emotional support that aligns more closely with the lived emotional rhythms of children with ASD.

Structuring Uncertainty Through Interaction Design. The structured dialogue in MOODMATE provided boundaries for emotional interaction without eliminating ambiguity. In the AR context, this structure was embodied through perceivable beginnings and endings of interaction, helping children orient themselves within emotional exchanges. Such structuring reduced cognitive load while allowing space for hesitation, vagueness, and non-linear responses. Here, structure functioned as a supportive mechanism rather than a normative guide.

Emotional Support as a Family-Level Practice. Parental involvement positioned emotional interaction as a family-level practice rather than an isolated technological intervention. Parents' role in contextual calibration enhanced the system's relevance to everyday life while also introducing variability across households. These findings suggest that emotional support systems for children should be understood as part of broader family practices, raising questions about how interpretive responsibility is shared between systems and caregivers.

6 Limitations and Future Work

This study adopted an exploratory deployment and qualitative analysis approach with a limited sample size. Accordingly, the findings are not intended to be generalized to all children with ASD. The current system remains at a prototype stage, and its ability to adapt emotional feedback to more complex or varied contexts requires further refinement. Future work may involve longer-term and more diverse deployments, as well as research designs that combine qualitative and quantitative methods, to further explore the potential of emotional interaction systems across different family contexts.

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